SGO in LLM On-Policy KD

Alephia 26/1/7

BACKGROUND

学生q_θ与教师o依据前缀w^{t-1}对下一个token w_t给出概率分布，然后做分布对齐

前缀来自教师o依据Prompt给出的Response(**TGO**, Teacher Generated Output)

或者来自学生q_θ依据Prompt给出的Response(**SGO**, Student Generated Output)

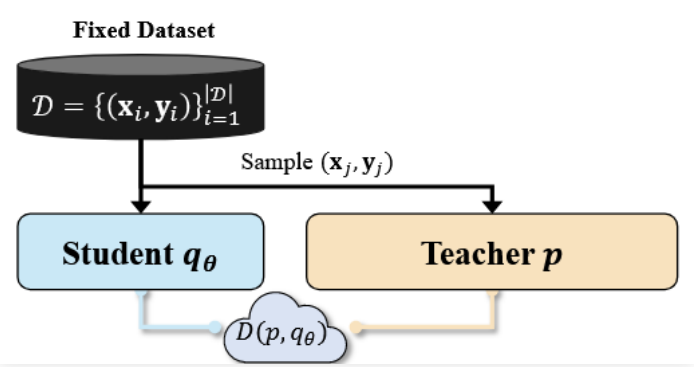
$$
\begin{aligned}
L(q_\theta; o) &= \sum_{t=1}^{T} \mathop{\mathbb{E}}_{w^{t-1}\sim d_o^{t-1}} \left[\mathop{\mathbb{E}}_{w_t\sim o(\cdot|w^{t-1})} \log \frac{o(w_t|w^{t-1})}{q_\theta(w_t|w^{t-1})} \right] \\
&= \sum_{t=1}^{T} \mathop{\mathbb{E}}_{w^{t-1}\sim d_o^{t-1}} \left[\mathcal{D}_{KL}(o(\cdot|w^{t-1})||q_\theta(\cdot|w^{t-1})) \right]
\end{aligned}
$$

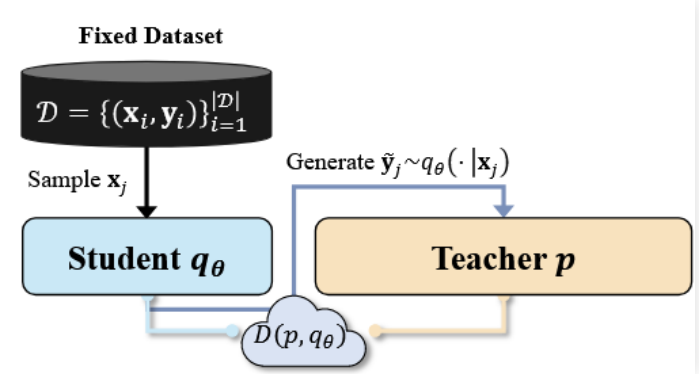
BACKGROUND

THE UTILIZATION OF SGOS(MAKING KD ON-POLICY)

$$L_{SGO}(q_\theta; o) = \sum_{t=1}^{T} \mathop{\mathbb{E}}_{w^{t-1}\sim d_{q_\theta}^{t-1},\, w_t\sim o(\cdot|w^{t-1})} \log \frac{o(w_t|w^{t-1})}{q_\theta(w_t|w^{t-1})}$$

每次有λ的概率使用SGO，$1-\lambda$的概率使用训练集样本





Agarwal, R., Vieillard, N., Zhou, Y., Stańczyk, P., Ramos, S., Geist, M., & Bachem, O. On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes. In ICLR, 24

BACKGROUND

TGOS AND SGOS IN LLM KD

🤔 单纯使用TGO，学生容易导致Train-Inference Mismatch

💡 使用SGO

🤔 学生生成的SGO，尤其在质量差时，相对目标分布以及教师容易OOD，教师难以给出有效指导，也没有训练意义

💡

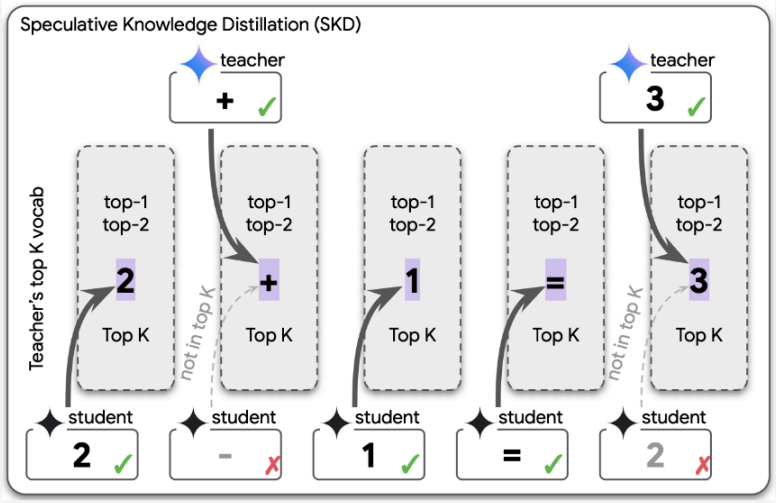
TGO, SGO混合使用

改进SGO(教师干预SGO生成)

SGO根据质量加权/低质量SGO反向对齐

SPECULATIVE KNOWLEDGE DISTILLATION

- 对于学生生成的token y_i，若其落在教师分布的Top-K，则接受，否则拒绝，由教师生成该token y_i^T
- 后续token丢弃，学生依据y_i^T生成后续SGO

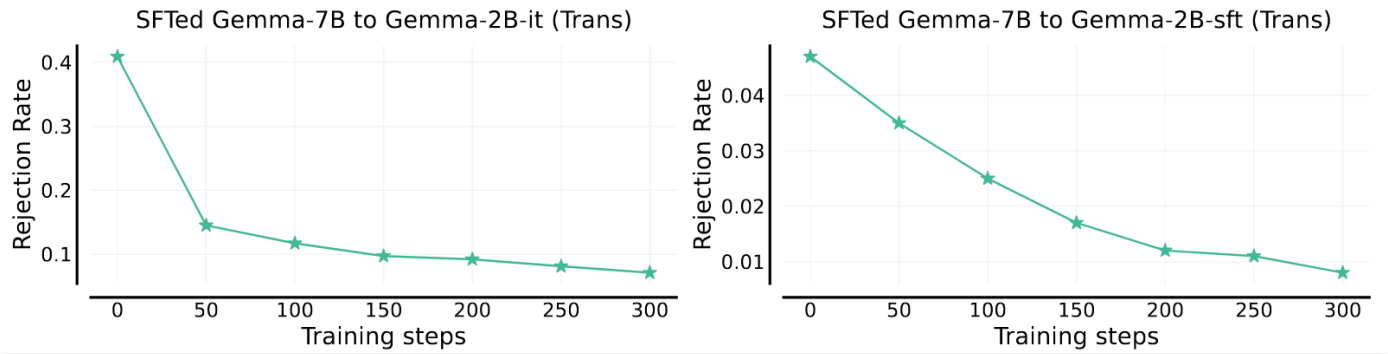


Xu, W., Han, R., Wang, Z., Le, L.T., Madeka, D., Li, L., Wang, W.Y., Agarwal, R., Lee, C., & Pfister, T. Speculative Knowledge Distillation: Bridging the Teacher-Student Gap Through Interleaved Sampling. In ICLR, 25

SPECULATIVE KNOWLEDGE DISTILLATION

前期偏向supervised knowledge distillation

后期偏向On-Policy knowledge distillation



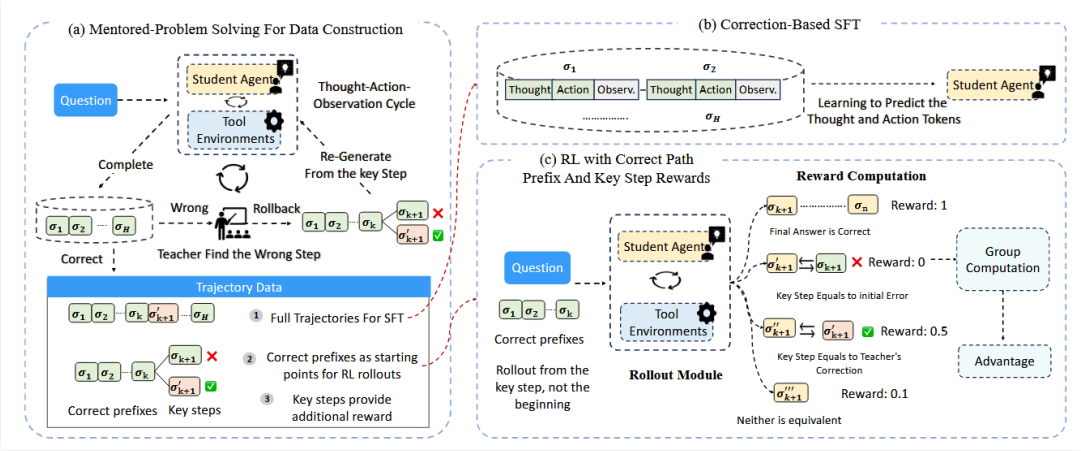
相比于纯TGO KD，计算量减小50%

SPECULATIVE KD IN INFERENCE

当SGO可以导向正确结果时，教师不进行纠正

否则找到第一个错误的token，修改，学生重新生成后续SGO

Too-Hard Samples will only be used in RL



Lyu, Y., Wang, C., Huang, J., & Xu, T. (2025). From Correction to Mastery: Reinforced Distillation of Large Language Model

SPECULATIVE KD IN INFERENCE

$$c(\pi) = \mathbb{E}_{s_1 \sim d^{\pi}, a_t \sim \pi(s_t), s_{t+1} \sim P_{\pi}(\cdot|a_t, s_t)} \left[\sum_{t=1}^{H} c_t(s_t) \right]$$

When $\mathbb{P}_{s \sim d_t^{\pi^T}}[\pi^S(s) \neq \pi^T] \leq \epsilon, \forall t \in 1, 2, \ldots H$, Then

$$c(\pi^S) \leq c(\pi^T) + O(H^2 \epsilon)$$

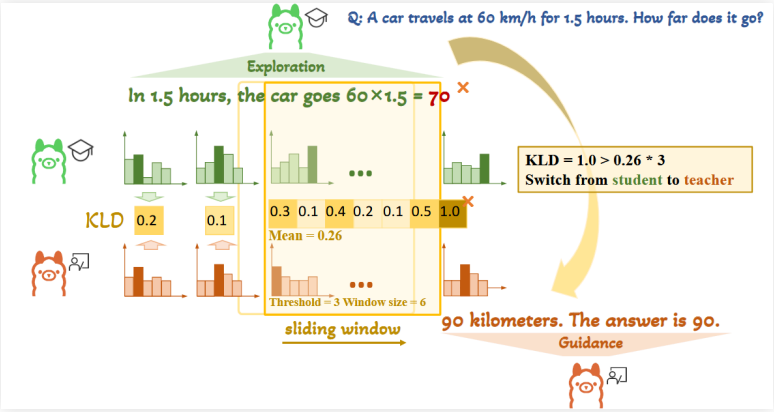
When $\mathbb{P}_{s \sim d_t^{\pi^{S'}}}[\pi^S(s) \neq \pi^{S'}] \leq \epsilon, \forall t \in 1, 2, \ldots H$, Then

$$c(\pi^S) \leq c(\pi^T) + O(H \epsilon)$$

ADAPTIVE SWITCHING GENERATION FOR KD

$$\overline{d_i} = \frac{1}{L} \sum_{j=i-L+1}^{i} \mathcal{D}_j(p||q_\theta), \text{ for } i \geq L, \text{ then } \tau_i = K \cdot \overline{d_{i-1}}$$

$$y_i \sim \begin{cases} q_\theta(\cdot \mid y_{<i}), & D_i(o||q_\theta) \leq \tau_i, \\ o(\cdot \mid y_{<i}), & D_i(o||q_\theta) > \tau_i \end{cases}$$



Peng, J., Wang, M., Cai, H., Li, Y., Zhang, K., Wang, S., Yin, D., & Zhao, X. (2025). AdaSwitch: Adaptive Switching Generation for Knowledge Distillation.

TEACHER INTERVENTED SGO

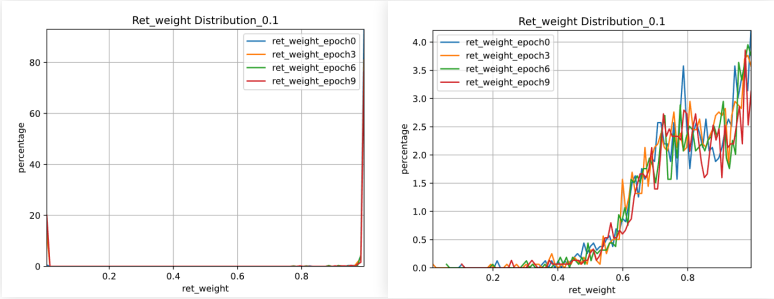
- Speculative Decoding
- Adaptive Switching Generation

缓解Train-Inference Mismatch

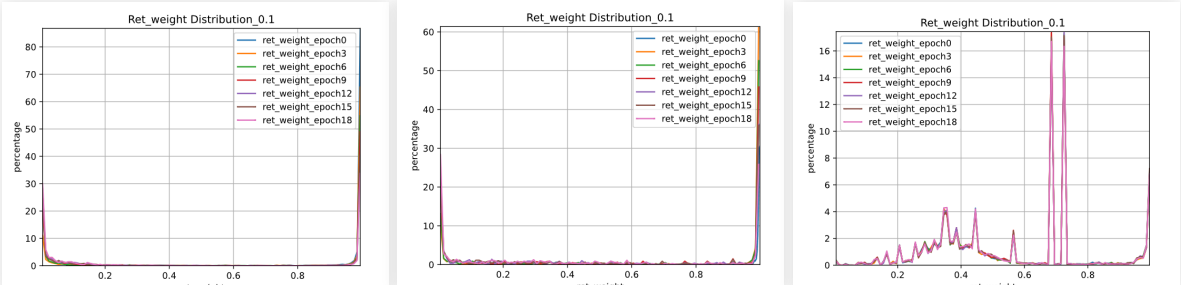
SGO-SPECFIC WEIGHTS

$$w_i = \frac{1}{Q(o; x_i)} = \exp \frac{1}{L} \sum_{j=1}^{L} \log o(x_i^j | x_i^{<j})$$

gpt2 on Dolly, Vicuna



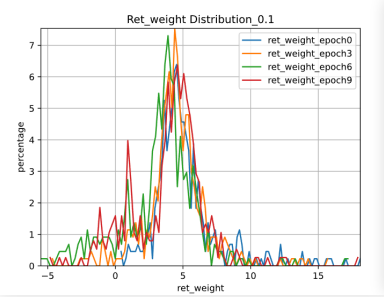
qwen2.5 on Dolly, Vicuna, Super-Natural

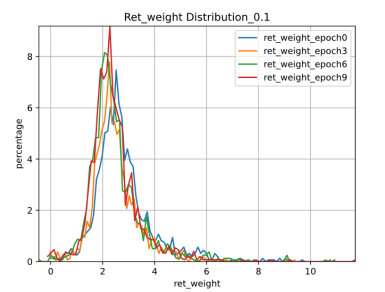


SGO-SPECFIC WEIGHTS

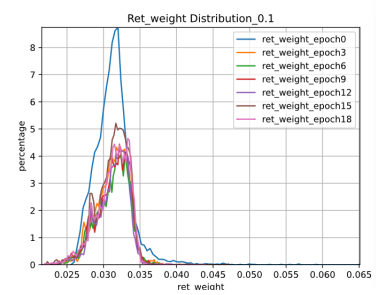
$$w_i = \exp\left(\frac{1}{L}\sum_{j=1}^{L}\log\frac{o(x_i^j|x_i^{<j})}{q_\theta(x_i^j|x_i^{<j})}\right)$$

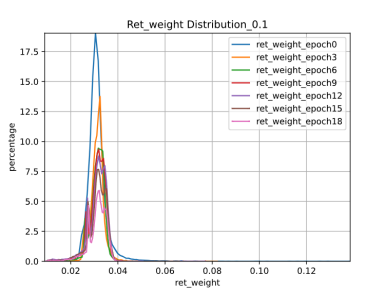
gpt2 on Dolly, Vicuna





qwen2.5 on Dolly, Vicuna





RESULTS

	gpt2-large(M_T) →gpt2-base(M_S)		qwen2.5-1.5B(M_T) →qwen2.5-0.5B(M_S)			Llama 3B(M_T) →Llama 1B(M_S)
	Dolly RougeL	**Vicuna RougeL**	**Dolly RougeL**	**Vicuna RougeL**	**Supernatural RougeL**	**Dolly RougeL**
M_T	27.7514	33.7246	30.6152	37.5300	36.5953	34.4764
On-Policy	**24.1342**	30.7272	**27.0046**	**37.5878**	36.0287	**32.6313**
/d, /w	22.4399	**31.7427**	26.6705	35.9461	**36.8834**	28.6582
/w	23.4788	31.2641	25.7262	36.8981	26.5557	32.2119
/d			25.7383		34.7877	29.2311
/w(T,S)			26.5996	35.9207	16.4174	32.0985

THE UTILIZATOIN OF LOW QUALITY SGOS

- 低质量SGO用于CE loss反向对齐
- 高质量SGO用于加权与教师对齐

$$\mathcal{L}_{\mathrm{PSR}}(\theta) = -\mathbb{E}_{\boldsymbol{x}\sim\mathcal{D}}\left[\sum_{\boldsymbol{y}:r(\boldsymbol{x},\boldsymbol{y})=1} \pi_\theta(\boldsymbol{y}|\boldsymbol{x})\right],$$

$$\mathcal{L}_{\mathrm{NSR}}(\theta) = -\mathbb{E}_{\boldsymbol{x}\sim\mathcal{D}}\left[\sum_{\boldsymbol{y}:r(\boldsymbol{x},\boldsymbol{y})=-1} -\pi_\theta(\boldsymbol{y}|\boldsymbol{x})\right].$$

直接对ce loss取反，不可行，后期会梯度爆炸

$$-\sum_{i=1}^{L} \log(q_\theta(x_i|x_{<i}))$$

$$L_{CALD} = \frac{1}{2|D|} \sum_{x,y_t,y_s\sim\mathcal{D}} \mathcal{D}_{SKL}(x, y_t) + \mathcal{D}_{SRKL}(x, y_s)$$

Ko, J., Chen, T., Kim, S., Ding, T., Liang, L., Zharkov, I., & Yun, S. (2025). DistiLLM-2: A Contrastive Approach Boosts the Distillation of LLMs. In ICML, 25

THANKS